Day 090 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Unit 6 Test

Standards: 6.EE.B.5, 6.EE.B.7, 6.RP.A.1, 6.RP.A.3a, 6.RP.A.2, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 3, 6, 11, 21

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enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

Directions

The first 4 questions (20%) review Unit 5. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Does x = 3 solve 2x + 7 = 11? Show substitution.

02 / Previous unit review | Constructed response

Solve x + 4 = 6. Check in the original equation.

03 / Previous unit review | Constructed response

Solve x - 8 = 5/2. Check in the original equation.

04 / Previous unit review | Constructed response

Solve 2 × x = 12. Check in the original equation.

05 / Constructed response

A tray has 7 red tiles and 2 blue tiles. Write the red-to-blue ratio in simplest form.

06 / Constructed response

A mix uses 6 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (6, 3), (12, ___), (36, ___).

07 / Constructed response

Mix A has 2 cups juice for 4 cups water. Mix B has 12 cups juice for 16 cups water. Which is stronger in juice? Show equivalent ratios.

08 / Constructed response

The ratio of oak trees to pine trees is 6:3. There are 45 trees altogether. Draw a tape diagram and find each count.

09 / Constructed response

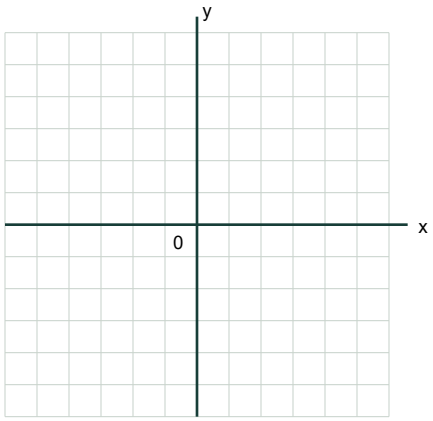
A mix uses 7 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (7, 3), (14, ___), (14, ___).

10 / Constructed response

Draw aligned number lines for 3 liters of paint covering 6 square meters. Mark 0, one batch, and 5 batches. How much area do 15 liters cover?

11 / Constructed response

For every 6 tickets sold, a club earns $3. Make a table for 0, 1, 2, and 5 batches. Plot (tickets, dollars), label axes, and interpret (30, 15).



Label the scale and units.

12 / Multiple select

3 notebooks cost $18. Select all statements at the same price per notebook.
A. 4 notebooks cost $24.
B. 4 notebooks cost $25.
C. Each notebook costs $54.
D. Each notebook costs $6.

13 / Two-part numeric entry and reasoning

Store A sells 5 notebooks for $20. Store B sells 6 notebooks for $30. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

14 / Multiple select

3 notebooks cost $6. Select all statements at the same price per notebook.
A. 2 notebooks cost $5.
B. 2 notebooks cost $4.
C. Each notebook costs $2.
D. Each notebook costs $18.

15 / Multiple select

A bicycle travels 120 km in 4 hours at constant speed. Select all true statements.
A. The speed is 124 km per hour.
B. The speed is 30 km per hour.
C. In 1 hour it travels 31 km.
D. In 3 hours it travels 90 km.

16 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 5/2 feet.
A. 31 inches
B. 5/24 inches
C. 30 inches
D. 5/2 × 12 inches

17 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 5/2 meters.
A. 1/40 centimeters
B. 5/2 × 100 centimeters
C. 250 centimeters
D. 251 centimeters

18 / Multiple select

A bicycle travels 75 km in 3 hours at constant speed. Select all true statements.
A. In 5 hours it travels 125 km.
B. The speed is 25 km per hour.
C. In 1 hour it travels 26 km.
D. The speed is 78 km per hour.

19 / Multiple select

A bag contains 4 red tiles and 10 blue tiles. Select all true statements.
A. There are 15 tiles in all.
B. The fraction of all tiles that are red is 2/7.
C. The red-to-blue ratio is 2:7.
D. The red-to-blue ratio is 2:5.

20 / Multiple select

A recipe uses 5 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (15, 6)
B. (10, 5)
C. (10, 4)
D. (21, 8)

Day 090 | Unit 06 | Answer key

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Standards: 6.EE.B.5, 6.EE.B.7, 6.RP.A.1, 6.RP.A.3a, 6.RP.A.2, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. No; left side = 13

2 × 3 + 7 = 13; compare with 11.

6.EE.B.5 | Constructed response

2. x = 2

Subtract 4 on both sides. Then x = 2; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

3. x = 21/2

Add 8 on both sides. Then x = 21/2; substitution gives 5/2 on each side.

6.EE.B.7 | Constructed response

4. x = 6

Divide by 2 on both sides. Then x = 6; substitution gives 12 on each side.

6.EE.B.7 | Constructed response

5. 7:2

Keep red first; divide 7:2 by 1.

6.RP.A.1 | Constructed response

6. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

7. Mix B

For 16 cups water, A uses 8 cups juice; B uses 12. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

8. 30 oaks and 15 pines

9 equal sections total; each section is 45 ÷ 9 = 5.

6.RP.A.3a | Constructed response

9. 6; 6

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

10. 30 square meters; aligned pairs (0,0), (3,6), (15,30)

15 ÷ 3 = 5 batches; 5 × 6 = 30.

6.RP.A.3a | Constructed response

11. (0,0), (6,3), (12,6), (30,15); 30 tickets earns $15

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

12. A, D - 4 notebooks cost $24.; Each notebook costs $6.

Cost per notebook = 18 ÷ 3 = 6. Multiply $6 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

13. A: $4 each; B: $5 each; A is the better buy.

Divide each cost by its own notebook count: 20/5 = 4; 30/6 = 5. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

14. B, C - 2 notebooks cost $4.; Each notebook costs $2.

Cost per notebook = 6 ÷ 3 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

15. B, D - The speed is 30 km per hour.; In 3 hours it travels 90 km.

Divide distance by time: 120 ÷ 4 = 30 km/h. Multiply this unit rate by 3 for the later distance.

6.RP.A.3b | Multiple select

16. C, D - 30 inches; 5/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 5/2 by 12, giving 30 inches. The physical length does not change.

6.RP.A.3d | Multiple select

17. B, C - 5/2 × 100 centimeters; 250 centimeters

There are 100 smaller units in each larger one. Multiply 5/2 by 100, giving 250 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

18. A, B - In 5 hours it travels 125 km.; The speed is 25 km per hour.

Divide distance by time: 75 ÷ 3 = 25 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

19. B, D - The fraction of all tiles that are red is 2/7.; The red-to-blue ratio is 2:5.

Divide both color counts by 2. For the fraction of the whole, use total 14 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

20. A, C - (15, 6); (10, 4)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select